PRETERM LABOR

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DEFINITIONS:

- Preterm labour (PTL) is the onset of labour before 37 weeks' gestation or presence of contractions which cause progressive effacement and dilatation of the cervix between 24 and 37 weeks' gestation.
- **Classification:**
- **Mildly preterm**: at 32+0 to 36+6 weeks
- (incidence *6.1%)*.
- **Very preterm:** at 28+0 to 31+6 weeks (incidence0*.9%)*.
- **Extremely preterm:** at 24+0 to 27+6 weeks (incidence *0.4%)*.

DEFINITIONS:

PROM: Premature rupture of membranes :

membrane rupture before the onset of uterine contractions(8% of term pregnancies)

PPROM: is the premature rupture of membranes before 37 completed weeks of gestation. It is about 3% and associated with, approximately one-third of PTL

Cervical insufficiency (formerly called cervical incompetence): It is painless cervical dilatation(mostly) resulting in delivery of a live fetus during the 2nd(early 3rd) trimester.

PTD CLASSIFICATION

PTD(preterm delivery),6-10 % of pregnancies can be categorized into:

A- spontaneous PTL 50%

B-PPROM 25%

C-delivery for maternal or fetal indications 25% Iatrogenic or medically-indicated deliveries are typically for diagnoses such as pre-eclampsia, fetal growth restriction (FGR) and maternal cardiac or renal conditions.

ETIOLOGY:

1- Prelabour rupture of membranes

2- Intra-amniotic infection (Chorioamnionitis)

3- Another ascending uterine infection (commonly due to group B streptococci)

4- Multiple pregnancy

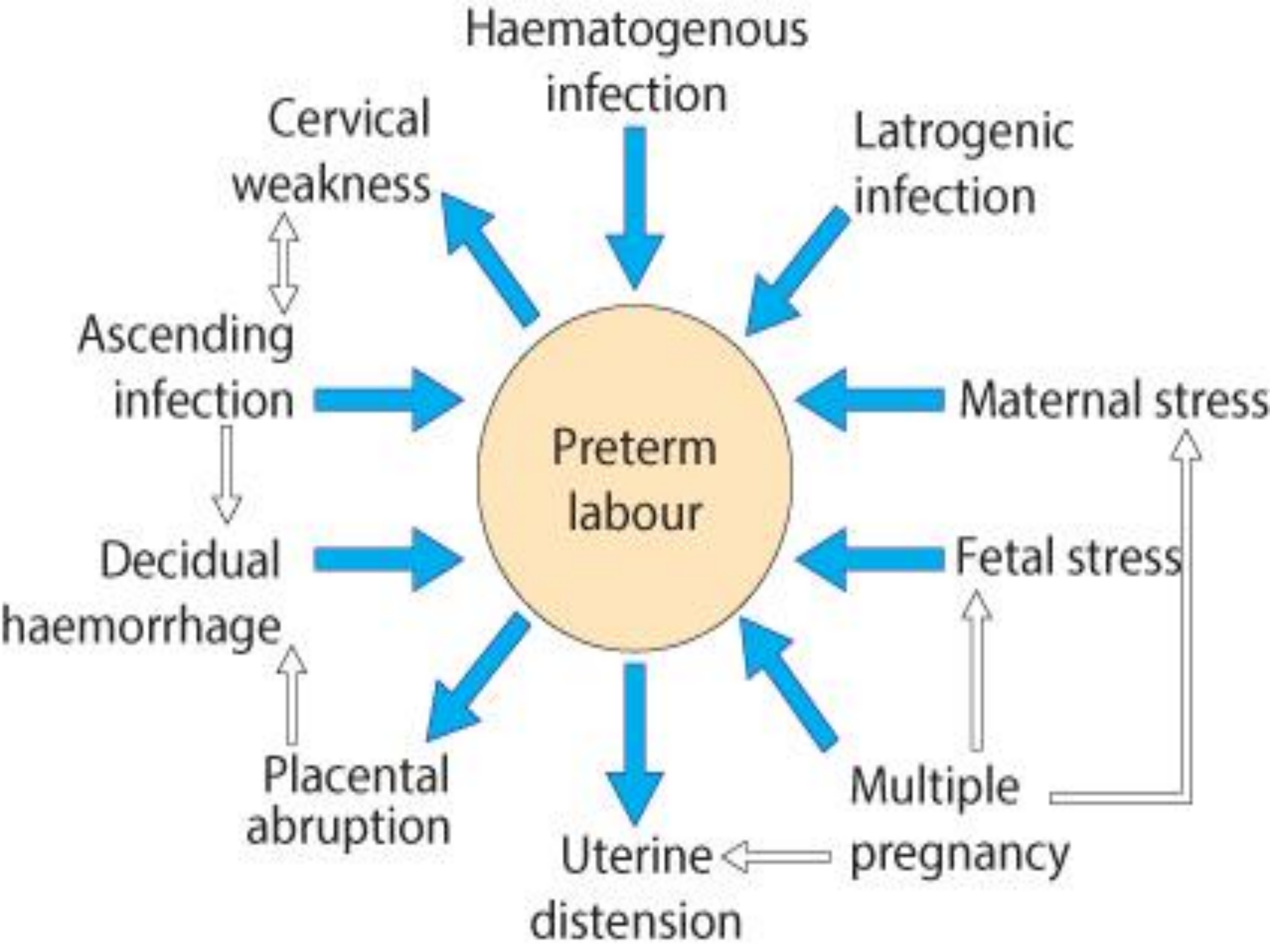
5- Fetal or placental abnormalities

6- Uterine abnormalities

7- Pyelonephritis

8- Some sexually transmitted infections (STIs)

9-idiopathic


Haematogenous infection
Latrogenic infection
Maternal stress
Fetal stress
Multiple pregnancy
Uterine distension
Placental abruption
Decidual haemorrhage
Ascending infection
Cervical weakness
Preterm labour

Risk factors for PTL/PPROM:

Non-modifiable, major

- One previous preterm birth: 20% risk.
- Two previous preterm birth: 40% risk.
- Twin pregnancy: 50% risk.
- Uterine abnormalities.
- Cervical anomalies like cervical damage
- Factors in current pregnancy like recurrent ante partum hemorrhage, illness and any invasive procedure or surgery.

RISK FACTORS FOR PTL/PPROM:

Non-modifiable, minor

- Teenagers having second or subsequent babies.
- Parity (0 or >5)
- Ethnicity (black women).
- Poor socioeconomic status.
- Education

RISK FACTORS FOR PTL/PPROM

Modifiable:

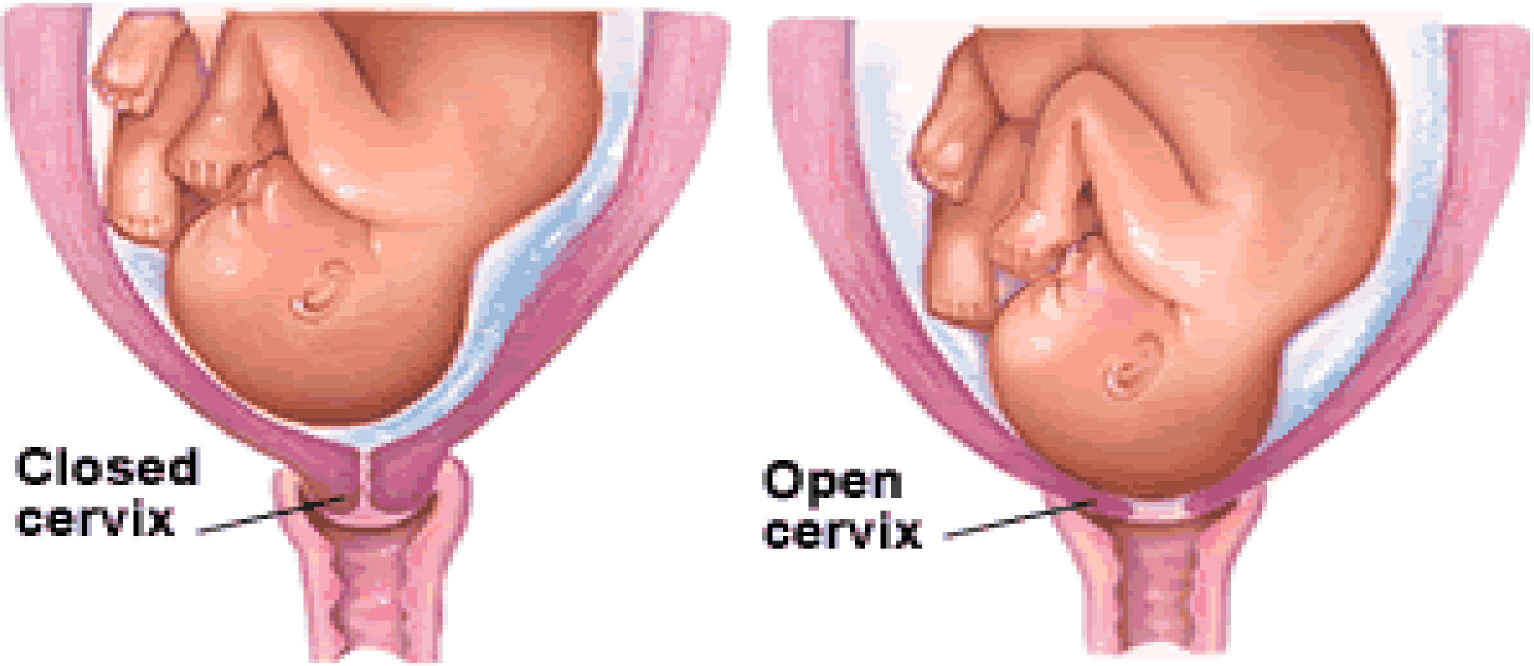
- Smoking: two-fold increase of PPROM.
- Drugs of abuse: cocaine.
- Body mass index (BMI) <20: underweight women.
- Inter-pregnancy interval<1 year.

CERVICAL WEAKNESS OR CERVICAL INSUFFICIENCY

painless cervical dilatation without regular contraction

Recently the features of cervical insufficiency is listed as follow:

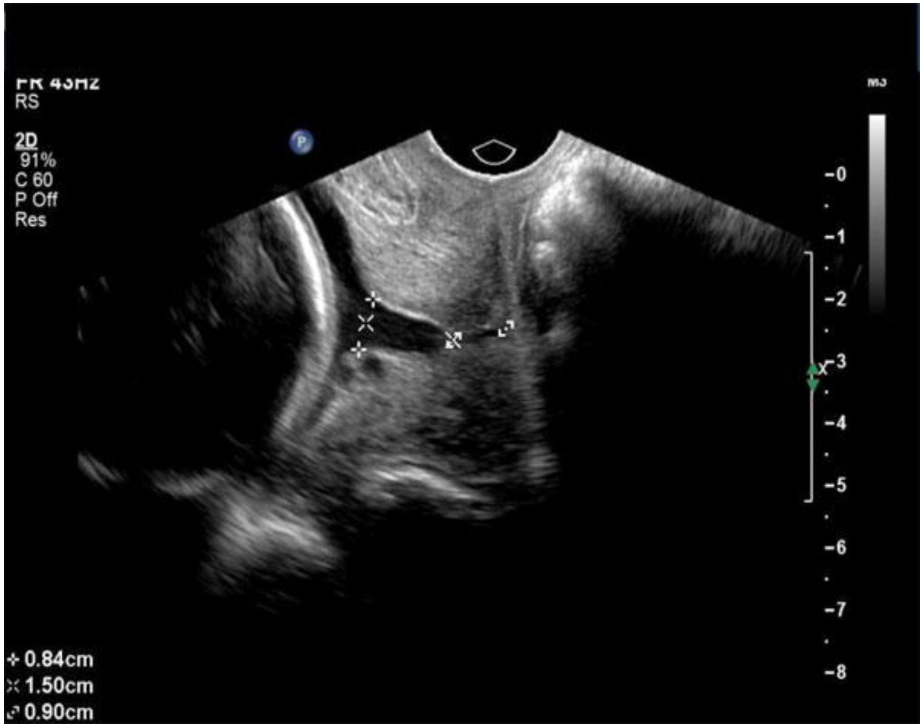
- History of two or more second trimester pregnancy losses
- History of spontaneous preterm birth prior to 34^{th} week of gestation
- History of painless cervical dilatation of up to 4-6cm.



- Absence of clinical finding consistence with placental abruption.
- History of cervical trauma caused by an operation or instrumentation on cervix e.g. cone biopsy, intrapartum cervical laceration or excessive forced cervical dilatation during curettage.

CERVICAL WEAKNESS OR CERVICAL INSUFFICIENCY

- The normal cervical length is about 3.5cm. Studies suggested that the risk of preterm birth is increased progressively as the cervical length decreased. At a cervical length of 26mm the risk increases 6-fold.



DIAGNOSIS/ PREDICTION AND DETECTION(MANAGEMENT)

- History and assessment of risk factors:

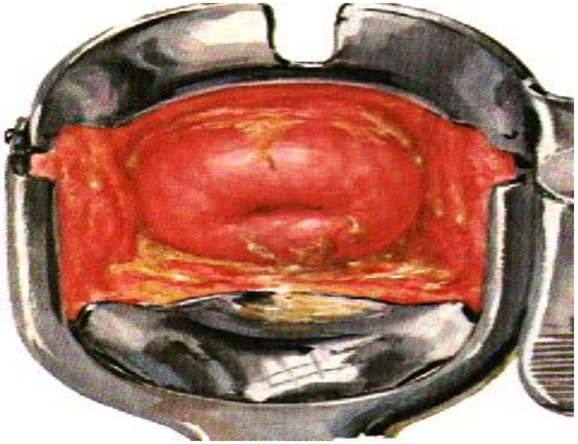
- In isolation, contraction is poorly correlates with the risk of preterm birth. Vague complaints such as increases discharge, pelvic pressure or low back ache are sometime reported, with latter two often showing a cyclical pattern.

EXAMINATION:

- A brief general examination. (pulse, blood pressure, temperature, and state of hydration).
- Abdominal examination may reveal the presence of uterine tenderness, suggesting abruption or chorioamnionitis.
- Digital examination should be limited, as they are known to stimulate prostaglandin production and may introduce organism into the cervical canal.

EXAMINATION

- A careful speculum examination by an experienced clinician may yield valuable information; pooling of amniotic fluid, blood and/or abnormal discharge may be observed. A visual assessment of cervical dilatation is usually possible and has been shown to be as accurate as digital examination finding.

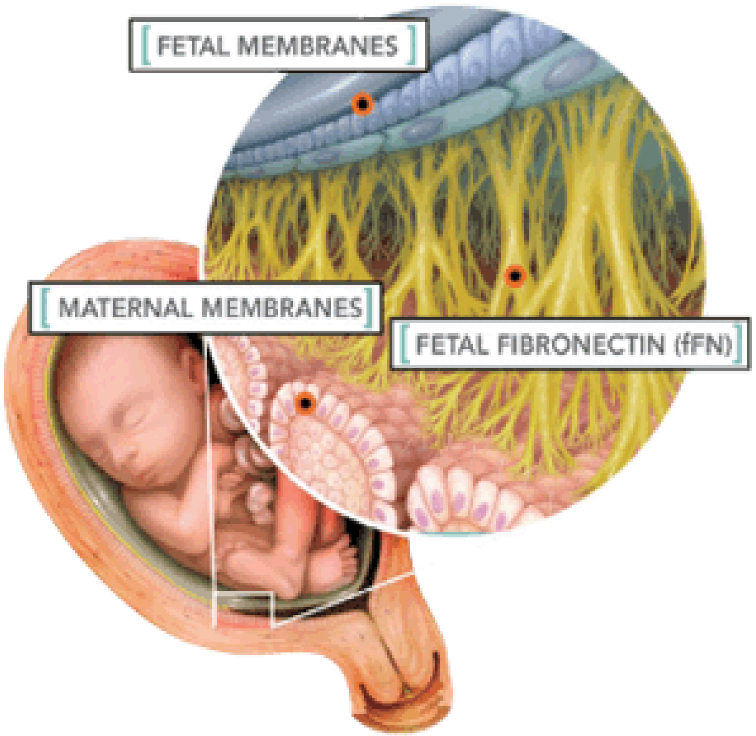


INVESTIGATIONS:

- <u>Bedside fibronectin testing:</u>

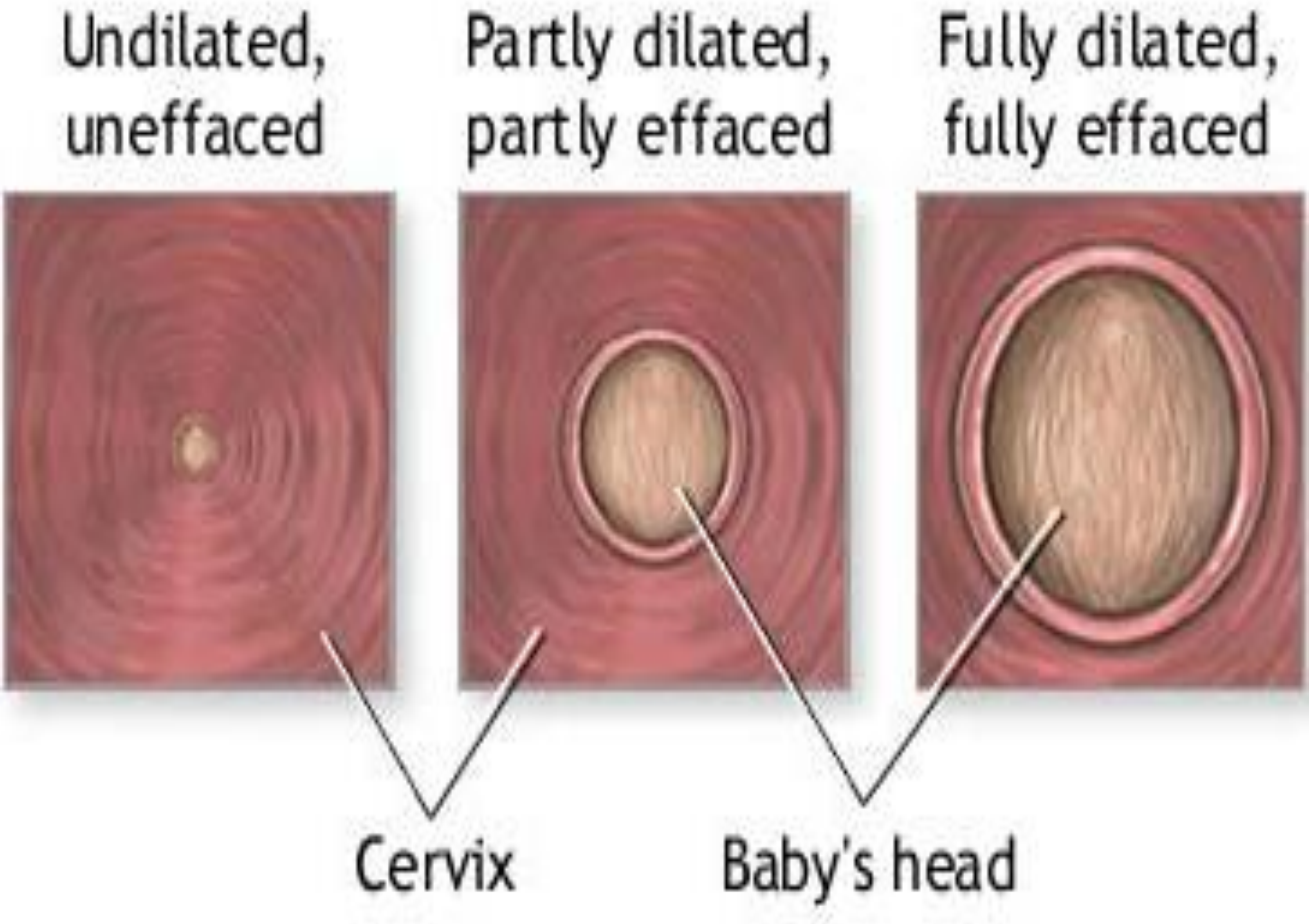
Fetal fibronectin (fFN) is a glue-like protein binding the choriodecidual membranes. It is rarely present in vaginal secretion between 23 and 34 weeks.

- In one study, 30% of women with a positive test delivered within 7 days, compared with only 10 percent of women who were 2-3 cm dilated.



INVESTIGATION

- Sonographic cervical length: It must be more than3.5 and no funneling, Significant cervical shortening is often accompanied by dilatation and funneling of the membrane down the cervical canal.
- Other investigations include CBP, ABO and Rh type, urine for culture and sensitivity and lower vaginal swab for culture and sensitivity
- Repeated vaginal examination: repeated vaginal examination in 1-4 hours should be considered essential in the absence of specialized tests.


Undilated,
uneffaced
Partly dilated,
partly effaced
Fully dilated,
fully effaced
Cervix
Baby's head

DIFFERENTIAL DIAGNOSIS:

- Urinary tract infection.
- Red degeneration of fibroid.
- Placental abruption.
- Constipation.
- Gastroenteritis.

Communication: two important areas of communication in management

- Communication with women and her family.
- Communication with neonatologist.

DRUG THERAPY:

- ***I-Steroids:***
- Current evidence shows that a single course of maternal steroids given between 26 and 34 weeks gestation and received within 7 days of delivery results in markedly improved neonatal outcomes, with significant reduction in rates of:
- Respiratory distress syndrome.
- Neonatal death.
- Intraventricular hemorrhage.
- Betamethason or dexamethason, better neurological outcomes with betamethasone.
- The dose is 12 mg two times only with 12-24 hours apart.

STEROIDS

- Maximum benefit from the injection is seen after 24 hours. Courses received less than 24 hours and more than 7 days before delivery did not produce a significant reduction in respiratory distress syndrome.
- Although there is reassuring evidence about the long term safety of steroid, repeated courses better to be avoided as it may causes:
- Increased risk of infection in PPROM.
- Restricted fetal body and brain growth.
- Adrenal suppression.
- Corticosteroid can cause significant glycaemic disruption in diabetic women.

DRUG THERAPY:

- ***II-Tocolysis:***
- The aim of tocolysis is to prolong gestation for at least until the administered corticosteroid will act which require 24hours to 7 days for maximum effect to be achieved and also to allow time for in utero transfer to a unit with suitable premature unit. However, tocolytic may be harmful because:
- It may prolong exposure of fetus to other wise harmful intrauterine environment like infection.
- No improvement in neonatal outcome.

CONDITIONS THAT PRECLUDE THE USE OF TOCOLYTIC:

- Progressive cervical dilatation.
- Chorioamnionitis.
- Fetal death or congenital anomalies.
- Bleeding (abruption placenta).
- Sever maternal medical or obstetrical conditions like sever maternal cardiac disease or sever pre-eclampsia.

TYPES OF TOCOLYTICS:

- *Beta-agonist:* like ritordine and salbutamol, these drugs only prevent delivery for 48 hours and not improve neonatal outcomes.
- side effects including, anxiety and palpitation or maternal death.

with greatest risk if it given:

- In large fluid volume.
- In multiple pregnancies.
- In women with cardiac diseases.
- In diabetic women additional care is needed as it causes significant extra glycaemic disruption.

DRUG THERAPY:

- *Oxytocin antagonist*: like atosiban, this is principally an Argenine vasopressin receptor antagonist but also binds the oxytocin receptors.
- *Other agents:* other smooth muscle relaxants used to treat preterm labour include magnesium sulphate, nifedipine, and GTN. There is little evidence to suggest increased efficacy or improved outcomes.
- Non-steroidal anti-inflammatory drugs (NSAIDs) such as indomethacine. Although it might delay preterm delivery for up to 7 days, but its undesirable drug because of its deleterious effects on renal ductus arteriosus and renal artery causing oligohydramnios.

DRUG THERAPY

III- Antibiotics : it does not seem to improve clinical outcome, but it may be beneficial in PPROM.

IV-Progesteron:

Current evidence points to progesterone being potentially beneficial in women at high risk of preterm delivery

MODE OF DELIVERY:

- Every effort should be made to transfer a women to a unit with facility of intensive preterm care unit.
- Appropriate analgesia is required to prevent un-necessary maternal effort.
- Appropriate fetal monitering.
- If gestational age is extremely low no need for C/S for fetal reason as neonatal out comes at this early gestations are very poor.

MODE OF DELIVERY

- As gestation advances, both neonatal outcome and ability to diagnose fetal compromise will improve, and intervention for fetal reason becomes universally appropriate.
- Until now the evidence suggest that preterm breech better to be delivered by C/S.

MANAGEMENT OF ASYMPTOMATIC HIGH-RISK WOMEN:

Investigation and treatment outside of pregnancy:

- Ideally, women should be seen postnatally and the events leading to their preterm birth reviewed.
- A careful analysis of events surrounding the previous birth should be undertaken in all patient who are at risk of having preterm birth.
- Following this, a clear management plan for any subsequent pregnancy should be made.

1- The importance of smoking cessation .

2- And, the potential benefit of leaving12 month pregnancies.

3- Dietician referral for women with a low BMI.

MANAGEMENT OF ASYMPTOMATIC HIGH-RISK WOMEN:

Investigation and treatment during pregnancy:

- **Early dating scan:** a first trimester dating scan is essential to time subsequent investigations.
- **Bacterial vaginosis** (BV): has been associated with an increased risk of preterm birth. The detection is by taking vaginal swab, if positive, then treatment is by giving oral metronidazole, which significantly lower the risk of preterm birth.
- **Asymptomatic bacteriuria:** this carries an increased risk of preterm birth; this may be due to increased risk of pyelonephritis.
- **Group B streptococcal colonization (GBS):** colonization has been linked to prematurity. Rectal and vaginal swab should be taken and if positive treatment should be given intrapartum.
- **Cervical ultrasound:** cervical length can be accurately and repeatedly measured by ultrasound. The risk of prematurity is inversely related to cervical length.

MANAGEMENT OF ASYMPTOMATIC HIGH-RISK WOMEN:

- Cervical cerclage: this indicated if the shortened cervical length is diagnosed by history, clinical and ultrasound finding. (I.e. cervical incompetence) treatment is applying suture at level of internal cervical os (McDonald versus Shirodkar cerclage).
- There is increasing evidence suggesting that **progesterone supplementation** can reduce the incidence of preterm delivery. This requires further evaluation.

NEONATAL OUTCOMES:

- **Survival:** predicted survival can be modified by accurate estimates of fetal weight or antenatal assessments of fetal well-being.
- **Morbidity:** preterm babies are at greater risk of:

1-Respiratory distress syndrome (RDS) and chronic lung diseases.

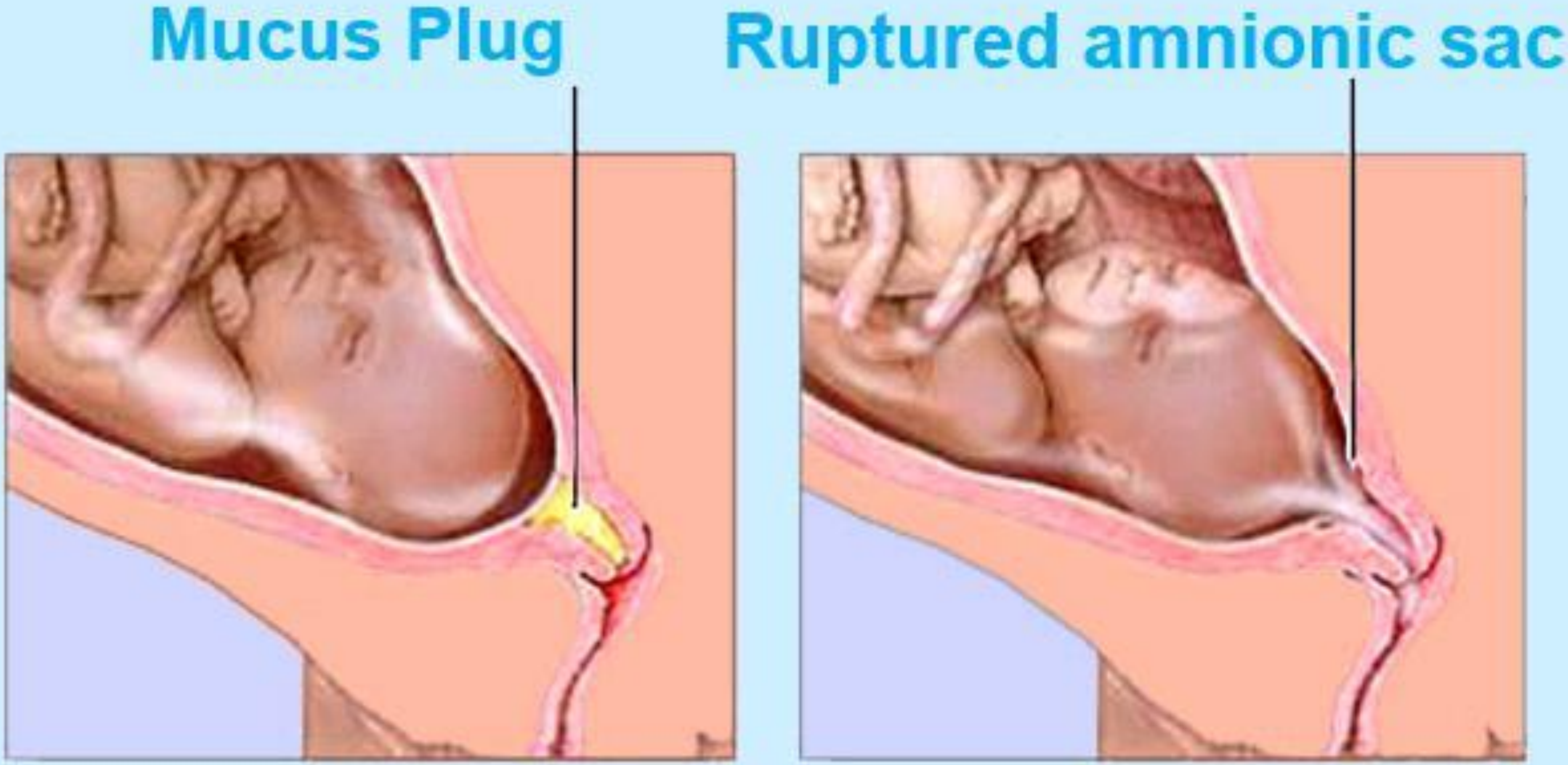
2-Cerebral damage.

3-Retinopathy.

4-Cerebral palsy.

5-Learning or hearing difficulty.

PROM AND PPROM



ETIOLOGY:

- ***Term PROM***
- The pathophysiology of PROM is not understood, but probably includes variety of mechanical, infective and constitutional mechanism.
- **Preterm PROM**

The risk factors for PPROM are similar to that of preterm labor and include:

- Ascending infection most important factor.
- Smoking which is dose dependant.
- Recurrent ante partum hemorrhage.
- Cervical weakness.

CLINICAL ASSESSMENT

History:

- A history from the mother of a gush of fluid followed by recurrent dampness will correctly identify over 90% of cases of pre-labor membrane rupture.

Examination:

- PROM should be confirmed by sterile speculum examination, performed after the mother has rested supine for 20-30minutes. Amniotic fluid can be seen pooling in the posterior fornix.
- A digital examination must be avoided unless in established labor, as it increase risk of chorioamnionitis, postpartum endometeritis and neonatal infection.

INVESTIGATIONS

- Bed side tests: the two most common test uses either nitrazine sticks (relying on the higher alkaline pH of amniotic fluid) or the ferning pattern seen when amniotic fluid is dried onto a glass slide and viewed under microscope
- Neither of these tests has been shown to more reliable than basic history and examination
- Vaginal specialized tests: these tests use technically expensive and advanced material by detecting substances that found in the amniotic fluid like BhCG, insulin like GFI, fFN and alpha microglobuline.
- Ultrasound: amniotic fluid volume can be assessed by ultrasound. Even at term there is variability in normal amniotic fluid volume making the usefulness of this test to be limited

CLINICAL MANAGEMENT

- factors linked with perinatal infection include:

- **An** increasing number of digital examinations after membrane rupture, **an** increasing interval between membrane rupture and labor onset, and **an** increasing duration of active labor.

- In general, the evidence suggests that immediate induction is associated with less maternal and neonatal infection and a shorter interval from membrane rupture to delivery than expectant management. There is no evidence that the mode of delivery is influenced and C/S is reserved for obstetric indications.

CLINICAL MANAGEMENT

- **Preterm PROM**
- The major risks with PPROM are:
- Chorioamnionitis.
- Abruption.
- Cord prolaps.
- Preterm delivery.
- Operative delivery.

- The patient should be admitted to hospital and managed as in-patient for 48 hours, as it is recognized that 48-72 hours is a critical time during which many women will labor or develop clinical chorioamnionitis. So it's recommended that all women with PPROM be managed as in-patient during this time. After that time the management is accordingly.

CLINICAL MANAGEMENT

Inside hospital:

1-Avoid digital examination and do sterile speculum examination.

2-The mother should be monitored for signs of chorioamnionitis by checking PR, temperature and uterine tenderness every 4 hourly.

3-The following investigation should be send :

- Complete blood picture for anemia and luekocytosis.
- C-reactive protein.
- Vaginal swab.
- Ultrasound for amniotic fluid volume and fetal well being.

4-Steroid: to reduce RDS and it does not increase chance of fetal or maternal infection.

5-Tocolytics: is relatively contraindicated in PPROM except until inutero transfer to a tertiary center and for steroid to act.

6-Antibiotic: this should be given in all cases of PPROM, and the best antibiotic is Erythromycin 250 mg qds for 10 days. If erythromycin is not available penicillin can be given except for amoxiclave as its associated with necrotizing enterocolits.

PREVIABLE PPROM BELOW 23-24 WEEKS

- Great dilemma as there is a critical complication of lung hypoplasia, limb abnormality, and extremely preterm birth with its morbidity and mortality.
- Many parents will opt for termination of pregnancy, although advanced technique has been used to seal the site of rupture without promising outcomes.

Chorioamnionitis

- The most morbid complication after PPROM for both the mother and the baby, and it's remaining notoriously difficult diagnosis. The criteria for diagnosis are as follow:
- Maternal pyrexia >38 C.

And at least two of either:

- Maternal tachycardia >100 bpm.
- Fetal tachycardia >160 bpm.
- Uterine tenderness.
- Raised C-reactive protein.
- Offensive vaginal discharge

CHORIOAMNIONITIS:

- When clinically suspected, delivery is almost always appropriate, as antibiotic therapy is rarely curative. The most common organism are anaerobes followed by group B streptococcus and then other streptococci.
- Apart from antibiotic the patient require analgesic, antipyretic and rehydration by IV fluid.


THANKS